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Maintenance and Optimization of Insertion Devices at NSLS-II using Motion Controls

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Outline

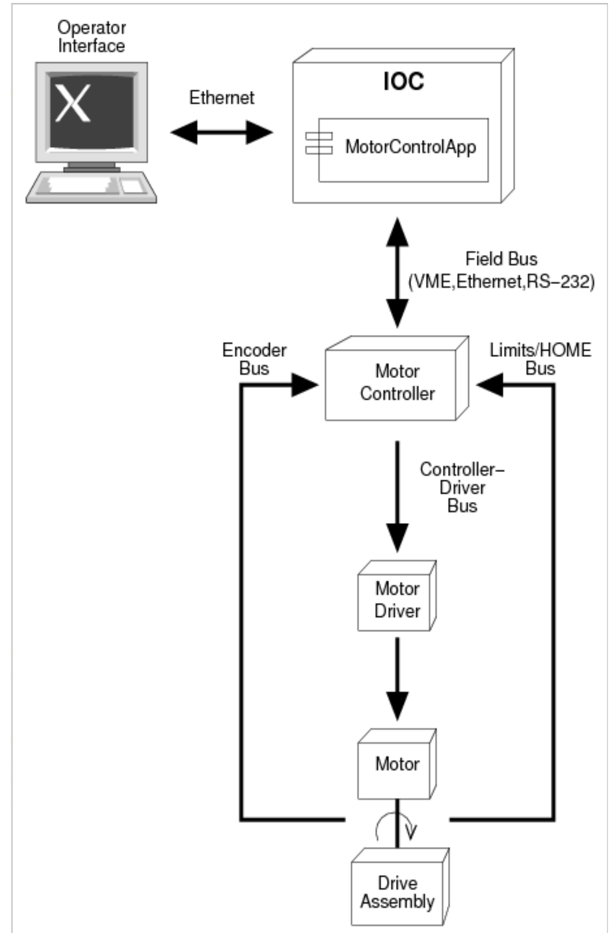
1. Preface on motion controls at NSLS-II
2. Accelerator motion controls
    1. Insertion devices (3PW, IVU, EPU, etc.)
    2. ID Issues
    3. ID Improvements
3. Q&A

Motion control hardware

- 64-bit Linux computers
- Delta Tau Programmable Multi Axis Controller (PMAC-2) and power supply modules
- Stepper/servo motors (and brakes where needed)
- Optical glass linear encoders
- Embedded rotary positional encoders (for some IDs)
- Ethernet, motor, encoder cables, etc.

Motion control software

- 3 Levels of software: controller configuration, IOC, and operator interface via EPICS communication
- Delta Tau Turbo PMAC GeoBrick LV is the standard motor controller at NSLS-II
- IOC has EPICS applications to channel access interface with motor controller
- User interfaces are OPI screens on CSS

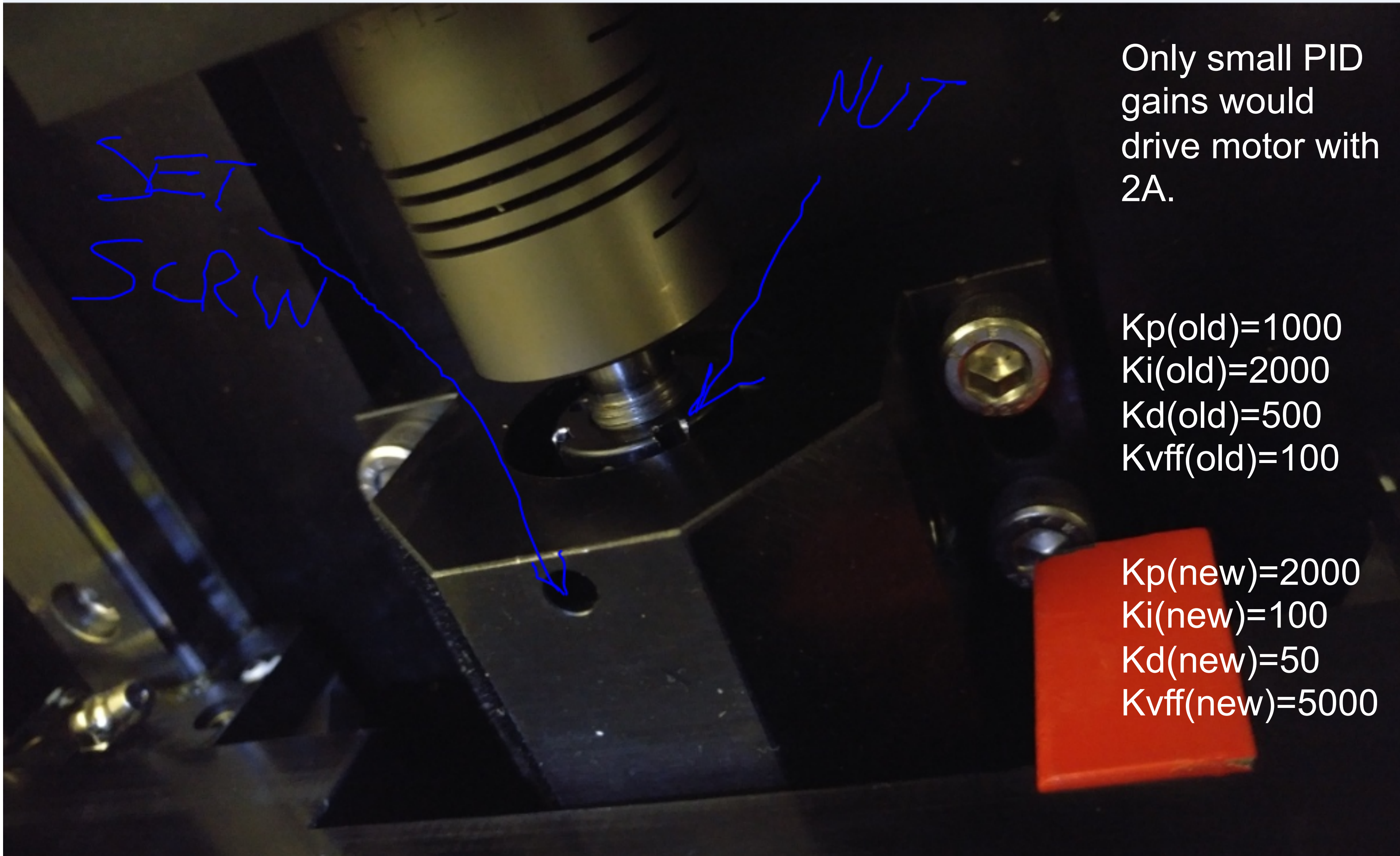


Insertion device issues

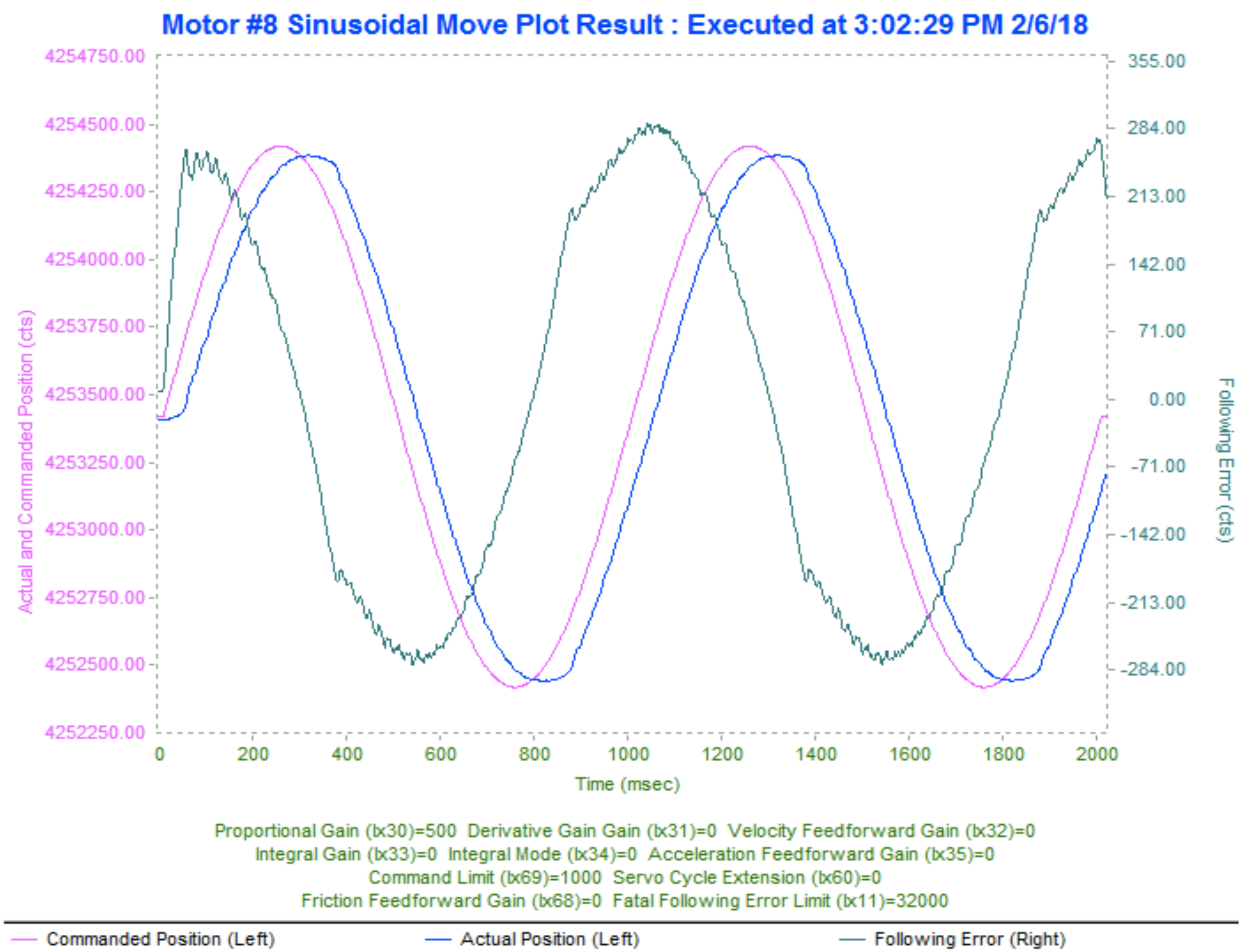
- 3PW motor moves too slow
- Undulator controls geometric issues
- EPU phase jumping
- Mechanical gear box/assemblies
- IOC server too slow

Proposed Solutions: cross couple gantry, retune PID/ adaptive tuning, ESA, migrate ID IOCs to new physical machine servers

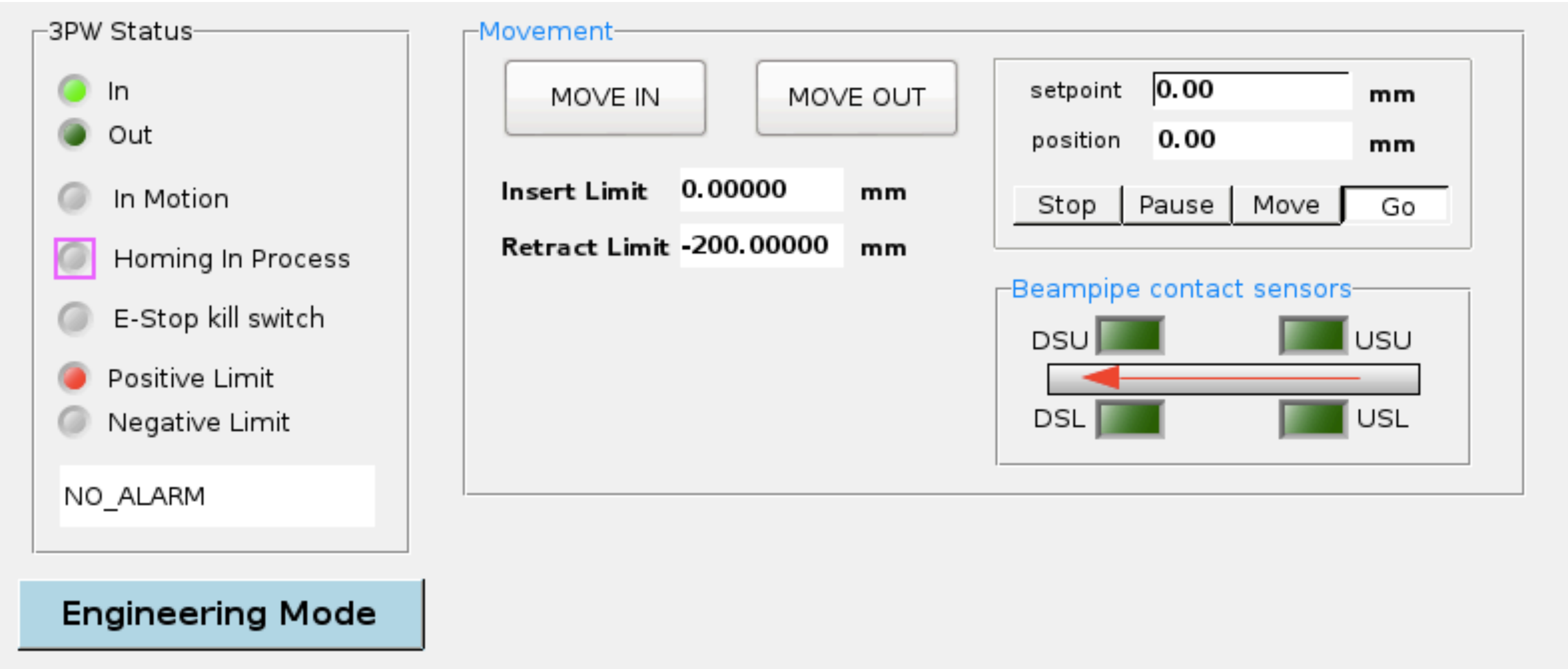
Loose 3PW Ballscrew



3PW Motion plot with loose ballscrew

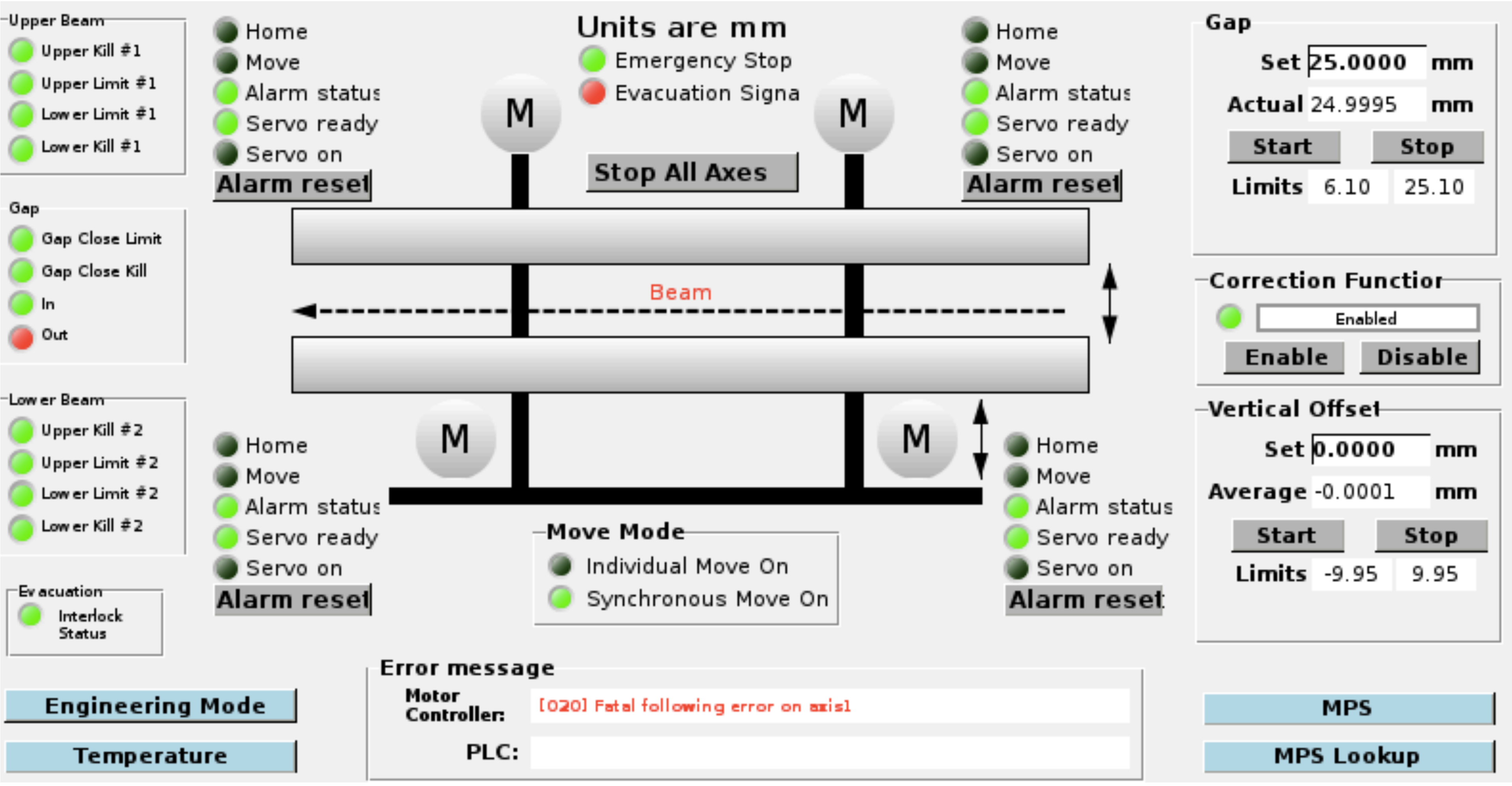


3PW CSS OPI Screen

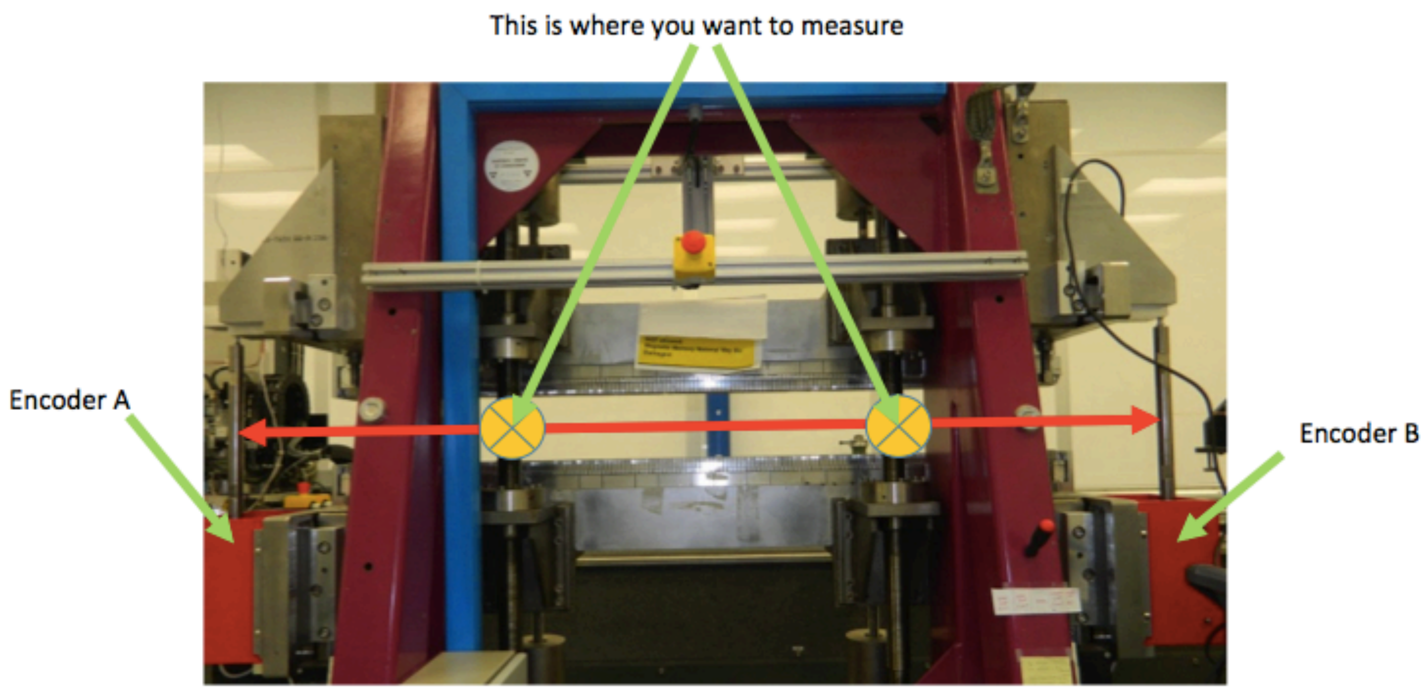


Was running at 0.2 mm/sec (15-20mins for full motion), now running at 2mm/sec

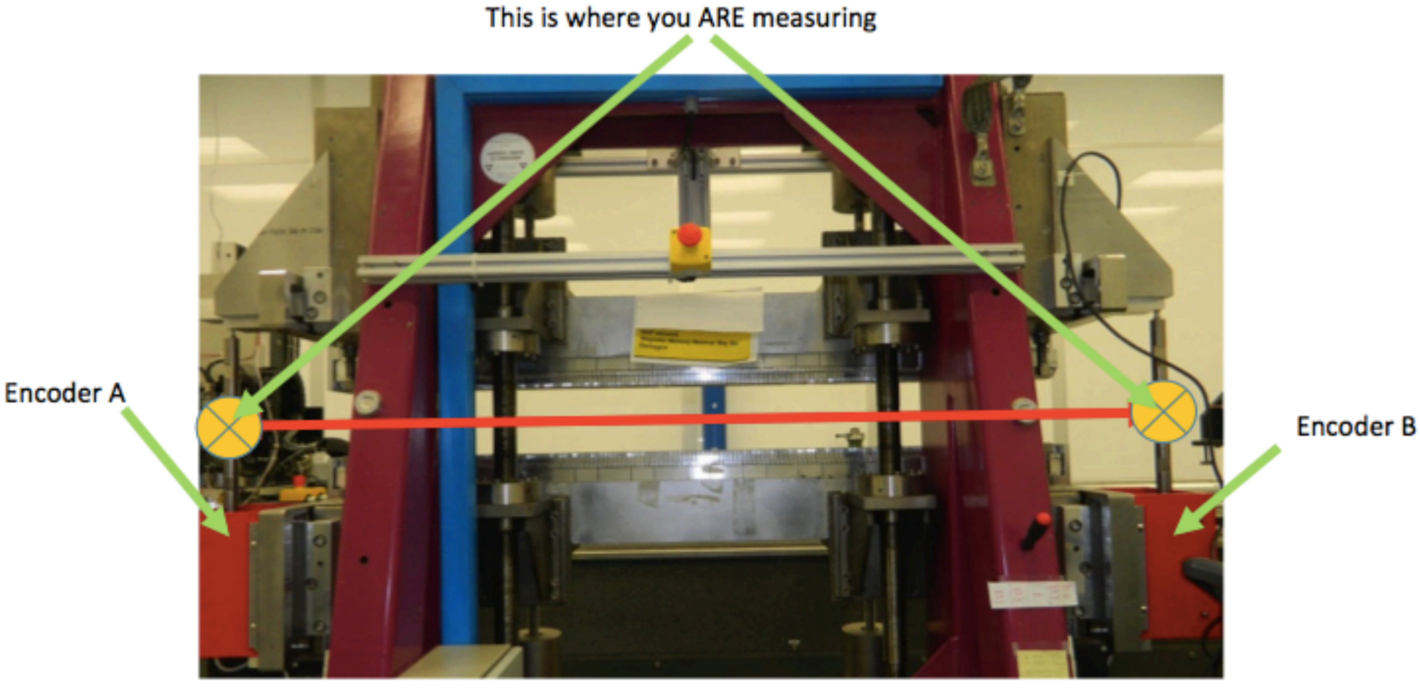
Neomax IVU CSS OPI screen



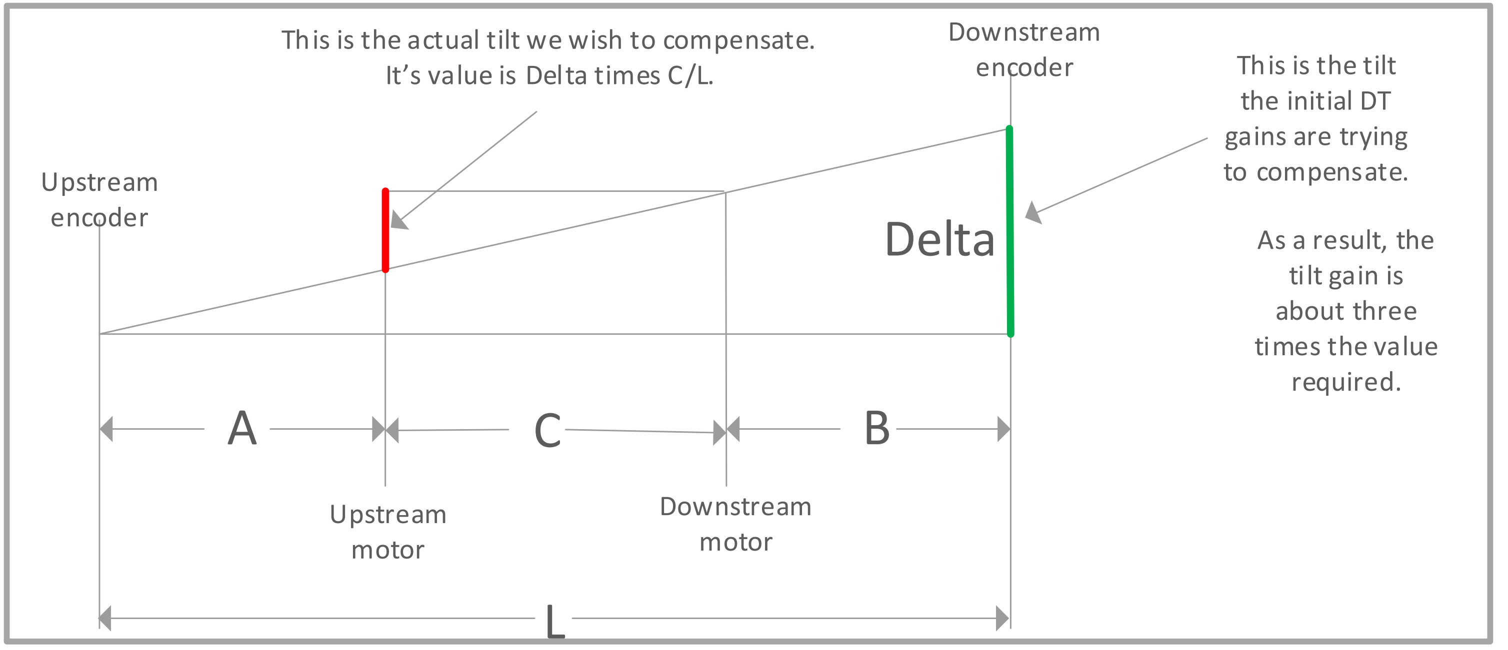
Undulator Cross-Coupled Gantry



Undulator Cross-Coupled Gantry



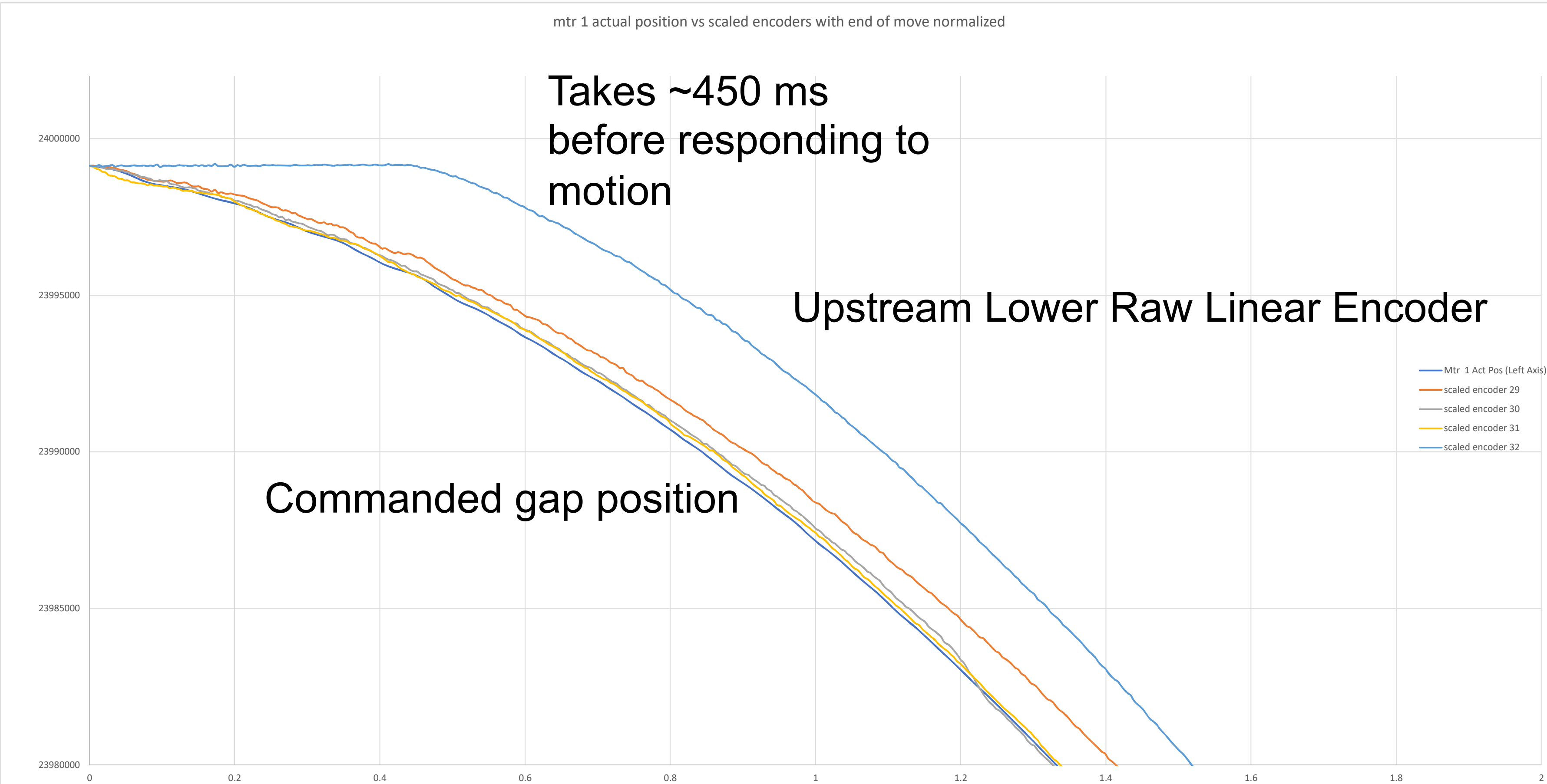
Geometry Cantilever Correction



- Give true ballscrew locations
- Remove positional error from cross couple gantry algorithm
- Allow higher proportional gain to reduce following error
- Eliminate need to install encoders directly on ballscrews

16-ID 3m IVU linear encoder data

Moving gap position from 24mm to 23 mm, 15 servo cycle gather period



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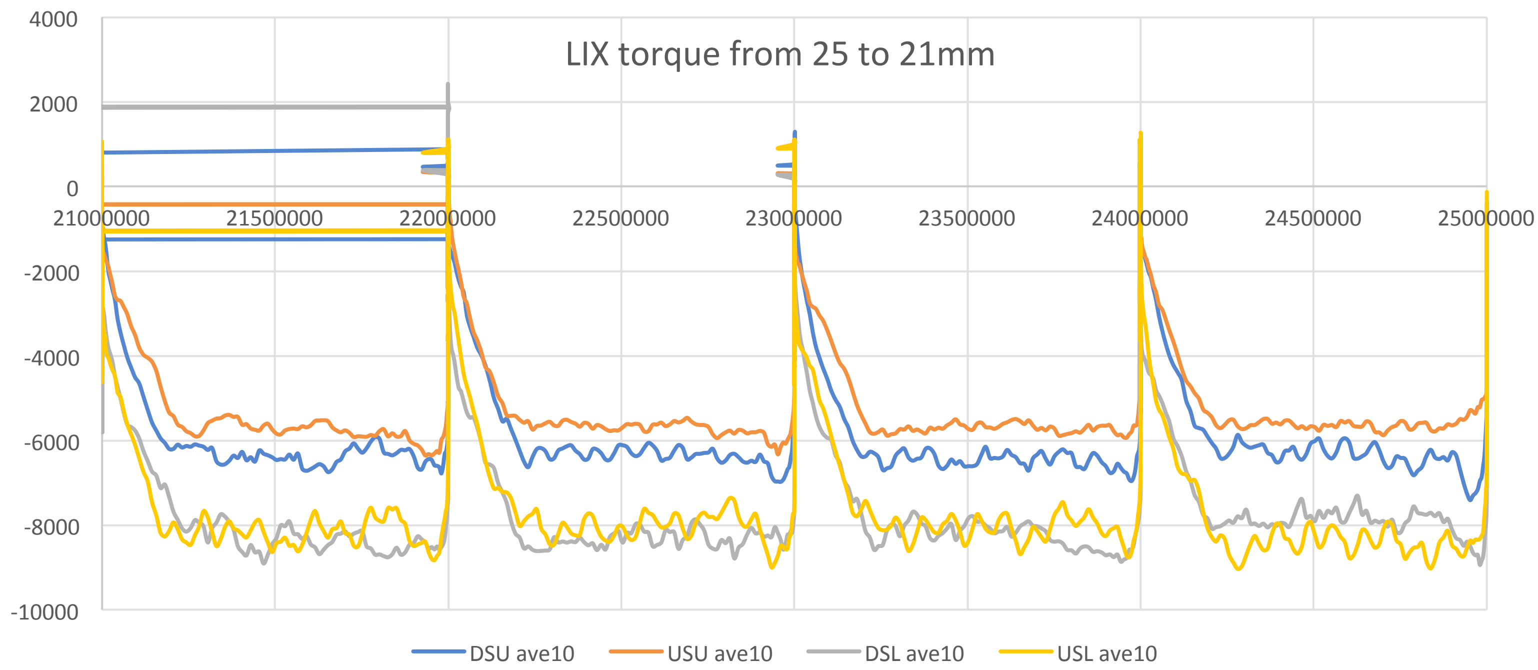
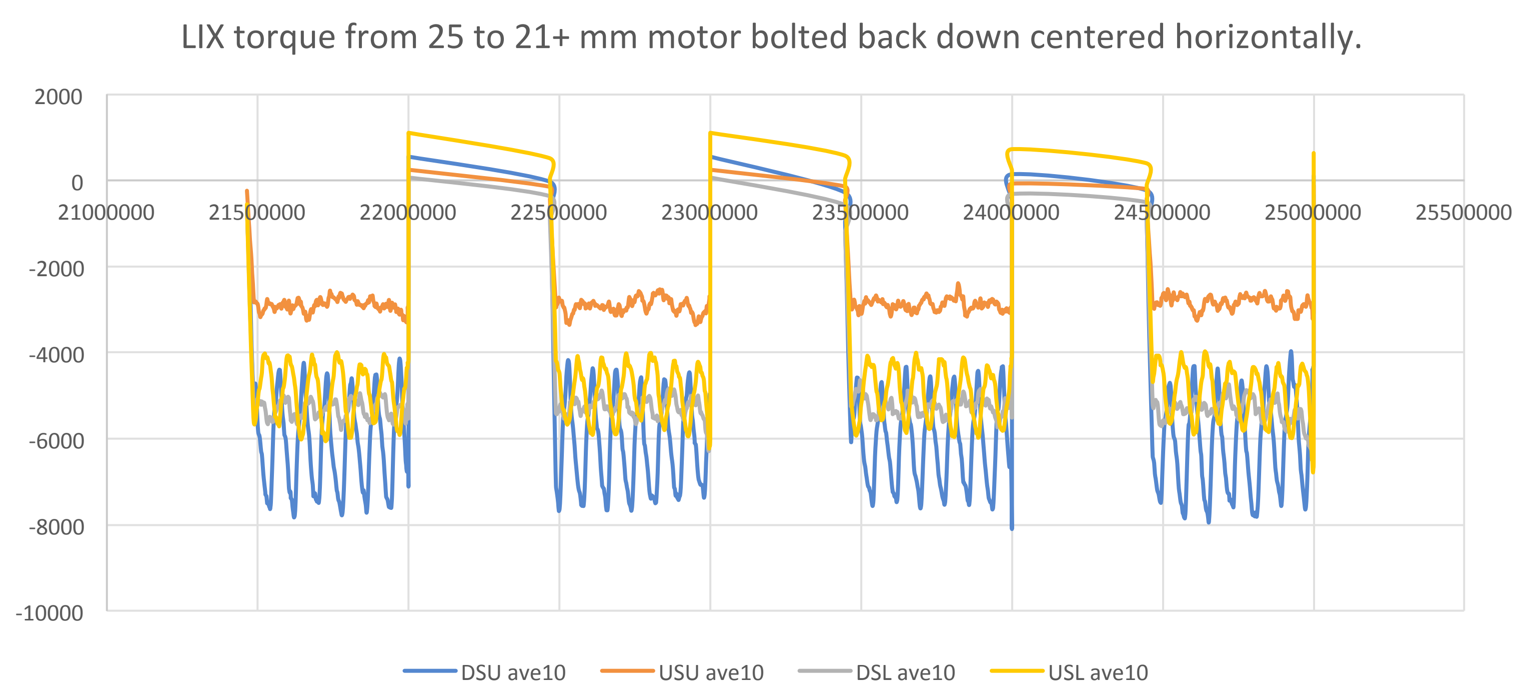
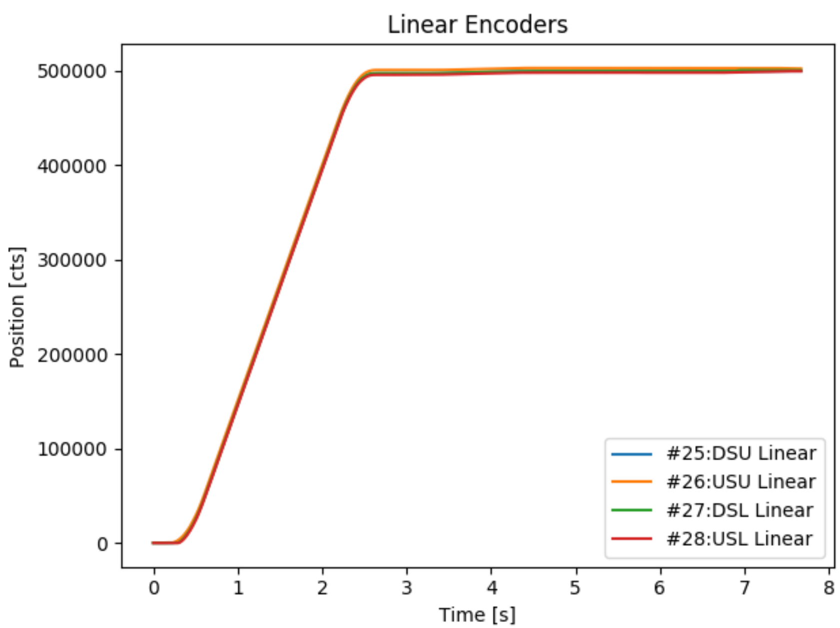
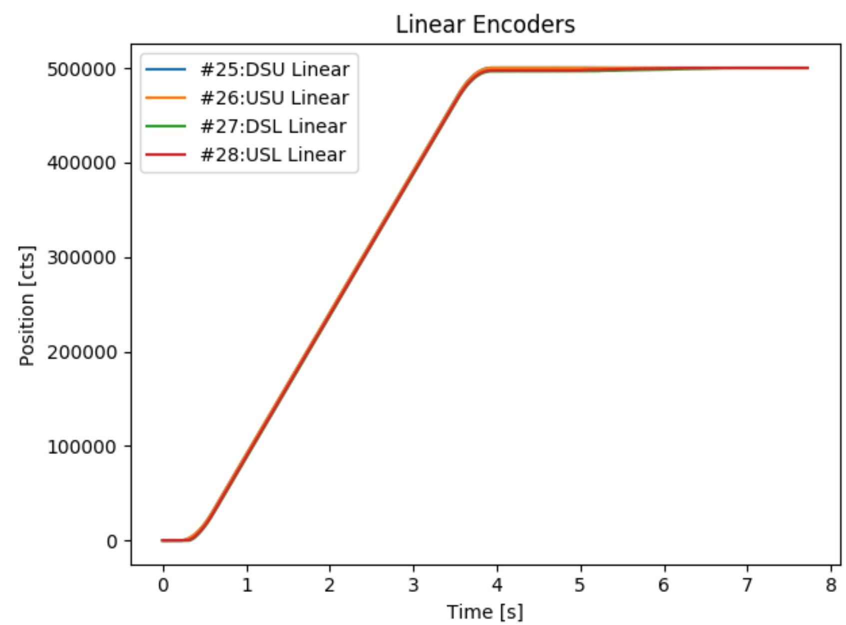

LIX torque from 25 to 21mm
4000
2000
0
-2000
-4000
-6000
-8000
-10000
21000000
21500000
22000000
22500000
23000000
23500000
24000000
24500000
25000000
DSU ave10
USU ave10
DSL ave10
USL ave10


LIX torque from 25 to 21+ mm motor bolted back down centered horizontally.
2000
0
-2000
-4000
-6000
-8000
-10000
21000000
21500000
22000000
22500000
23000000
23500000
24000000
24500000
25000000
25500000
DSU ave10
USU ave10
DSL ave10
USL ave10

Repeatability of IVU motion

Starting gap at 25mm ending at 24mm





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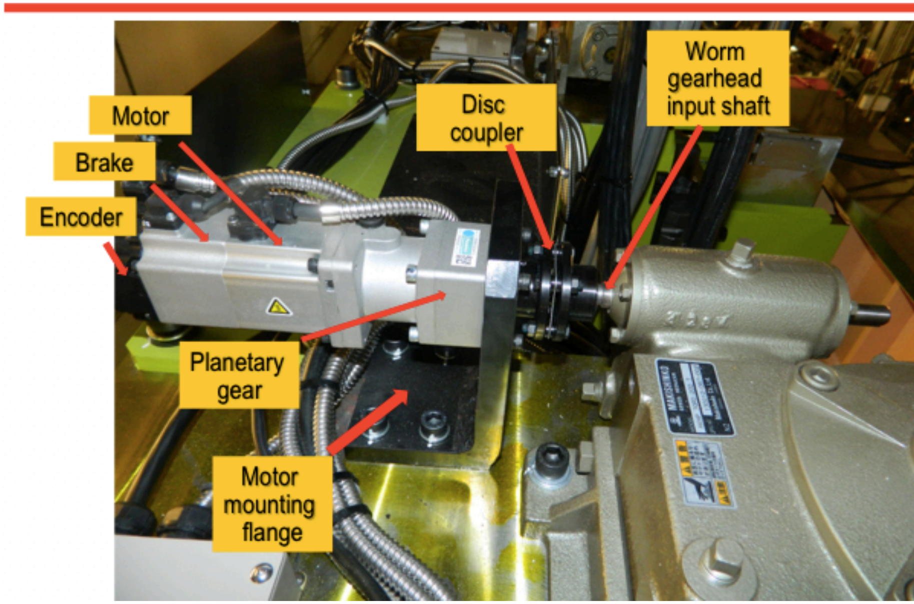
Currently under development

- Redesign of TPMAC source code for Neomax IVU at cells 4,12, and 16
- Mechanical repairs on undulators drive train
- EPU Phasing

1.5 meter IVU Drive train coupling



Acknowledgements

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Any Questions?